Tuesday, May 26, 2026 8:29 PM

CNN — Convolutional Neural Network

→ Deep learning Algorithm

→ Especially used for image training & classification

Till now we handled STRUCTURED DATA

↓

tabular data / numerical data

(ML, ANN)

Unstructured data (images, text, video, audio)

↓

CNN

IMAGE

Image is a 2D representation of real world objects/entities made up of PIXELS

0 – 255

GRAYSCALE

GRAYSCALE

R G B

Black (0, 0, 0)

White (255, 255, 255)

RGB-image

	R	G	B
R	255	0	0
G	0	255	0
B	0	0	255

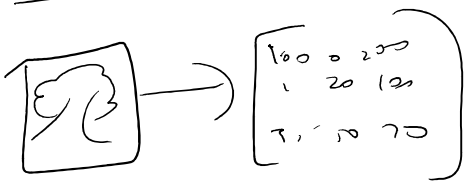
Different layers in CNN

→ CONVOLUTIONAL LAYER

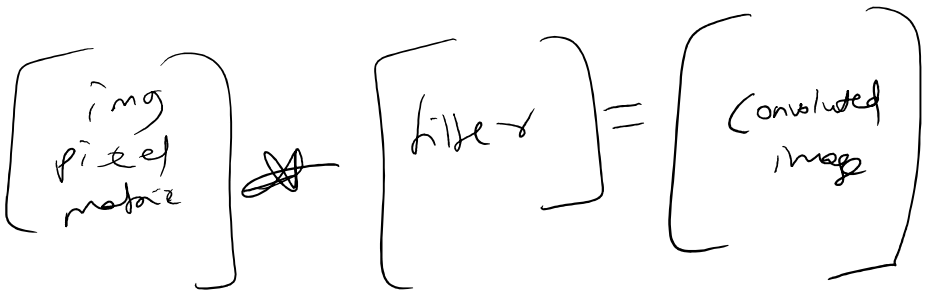
→ POOLING LAYER

→ FULLY CONNECTED LAYER

Convolution layers



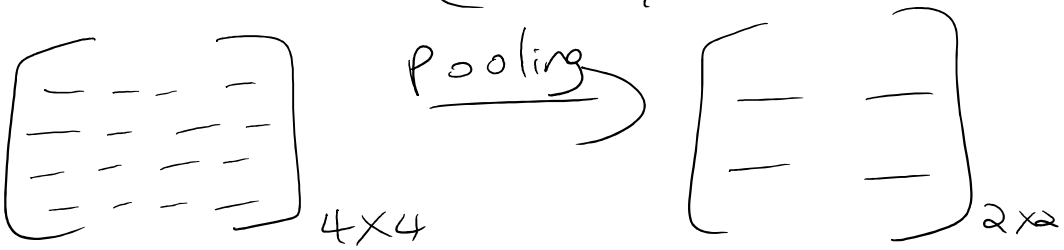
$$\begin{bmatrix} 100 & 0 & 200 \\ 5 & 20 & 120 \\ 3 & 70 & 70 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} - & - \\ - & - \end{bmatrix}$$



Convolution layers perform convolution operation of the image with a set of filters to produce feature maps & we will be getting a Convoluted image as o/p.

POOLING LAYER

* Process of downsampling the image (compressing/reducing the size)



16X16 → 8X8
8X8 → 4X4
4X4 → 2X2

FULLY CONNECTED LAYER

↓

→ Combination of i/p, hidden, o/p

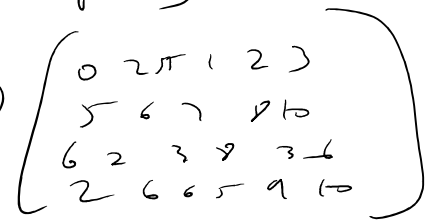
→ Layer used for training the image

CNN → Conv layers + Pooling Layer + ANN

Step-by-Step Working of CNN

Step 1 :- Take the image 

Step 2 :- Make the img to pixels

 → 

Step 3 :- Choose a filter

Step 4 :- Perform convolution o/p

[img] * [filter] = [C.V]

Step 5 :- Apply Pooling layer

Step 5:- Apply pooling layer

[C.V] $n*n$ → Pooling → [Pooled ing] $n/2 * n/2$

Step 6:- Repeat Convolution & Pooling until you get the most standardized image

Step 7:- Flatten the matrix
(Converting 2d to 1d for giving to fully connected layer)

Step 8 !— Pass it into fully connected layer for training

TYPES OF FILTERS

filters in CNN is used to extract specific features from our image
Such as the edges, the centre.

- PREWITT FILTERS

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

- SOBEL FILTERS

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

- LAPLACIAN FILTERS

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

TYPES OF POOLING

Max Pooling

Min Pooling

Average Pooling



→ Max → $\begin{bmatrix} 9 & 8 \\ 9 & 7 \end{bmatrix}_{2*2}$

→ Min → $\begin{bmatrix} 2 & 3 \\ 0 & 6 \end{bmatrix}_{2*2}$

4*4

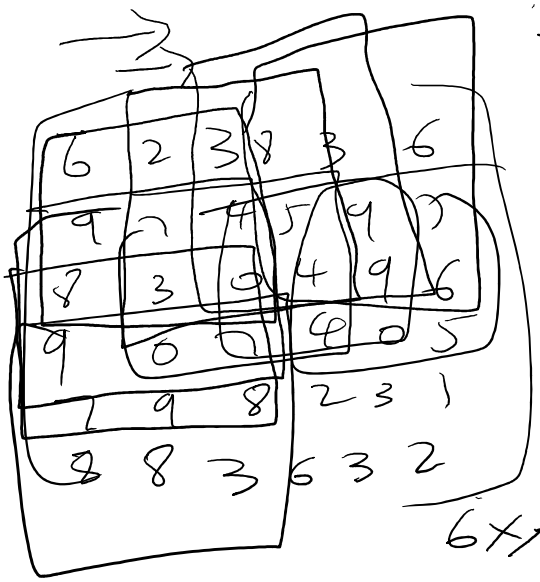
Avg → $\begin{bmatrix} 6 & 5 \\ 5 & 4 \end{bmatrix}_{2*2}$

Mathematical Implementation / Example

$$\begin{bmatrix} 6 & 2 & 3 & 8 & 3 & 6 \\ 9 & 7 & 4 & 5 & 9 & 7 \\ 8 & 3 & 0 & 4 & 9 & 6 \end{bmatrix} \rightarrow \text{IMG}$$

$$\begin{bmatrix} 8 & 3 & 0 & 4 & 9 & 6 \\ 9 & 0 & 7 & 4 & 0 & 5 \\ 7 & 9 & 8 & 2 & 3 & 1 \\ 8 & 8 & 3 & 6 & 3 & 2 \end{bmatrix}_{6\times 6}$$

PIXEL



$\rightarrow$ STRIDE 1

$$\circledast \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}_{3\times 3}$$

$$= \begin{bmatrix} 21 & -3 & -19 & -4 \\ 15 & 8 & 3 & 6 \\ 9 & 8 & 7 & 6 \\ 20 & -6 & 9 & 16 \end{bmatrix}_{4\times 4}$$

$$\begin{bmatrix} 6 & 2 & 3 \\ 9 & 7 & 4 \\ 8 & 3 & 0 \end{bmatrix} \circledast \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} =$$

$$6\circledast 1 + 2\circledast 0 + 3\circledast -1$$
$$+ 9\circledast 2 + 7\circledast 0 + 4\circledast -2$$
$$+ 8\circledast 1 + 3\circledast 0 + 0\circledast -1$$

$$= 6 + -3 + 18 + -8 + 8 = 21$$

$$\begin{bmatrix} 2 & 3 & 8 \\ 7 & 4 & 5 \\ 3 & 0 & 4 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = 21$$

$= 2*1 + 8*-1 + 7*2 + 5*-2 + 3*1 + 4*-1$

$= 2 + -8 + 14 + -10 + 3 + -4$

$= -3$

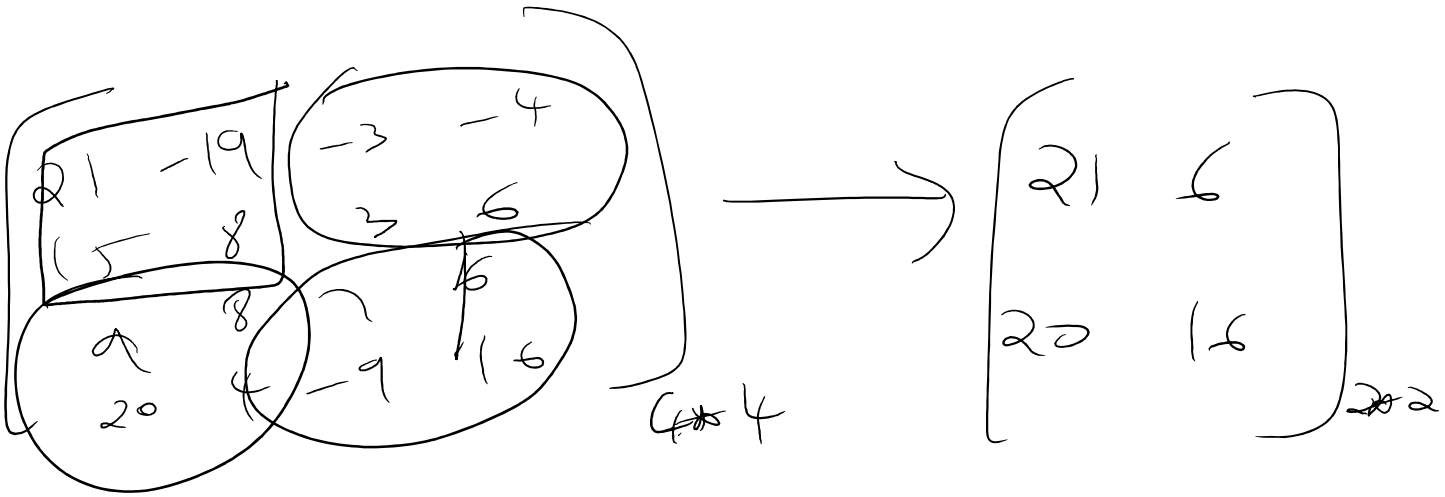
$$\begin{bmatrix} 3 & 8 & 3 \\ 4 & 5 & 9 \\ 0 & 4 & 9 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = 3*1 + 3*-1 + 4*2 + 9*-2 + 9*-1$$

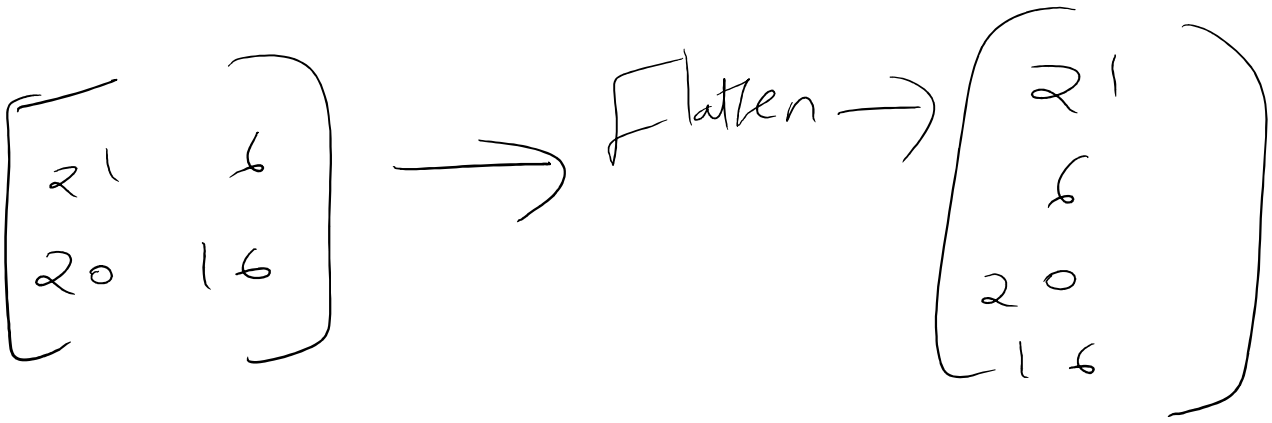
$= 8 + -18 + -9$

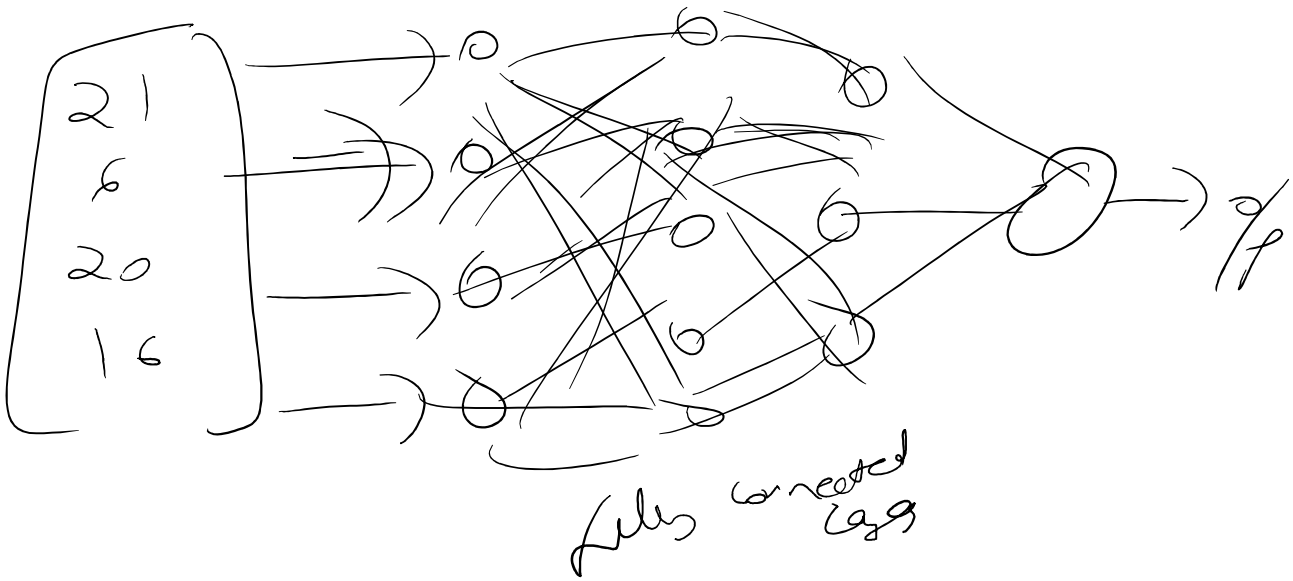
$= 8 + -27 = -19$

$$\begin{bmatrix} 8 & 3 & 6 \\ 5 & 9 & 7 \\ 4 & 3 & 6 \end{bmatrix} * \begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix} = 8*1 + 6*-1 + 5*2 + 7*-2 + 4*1 + 6*-1 = 8 + -6 + 10 + -14 + 4 + -6$$

$= -4$


21 -19
15 8
20 4
-3 -4
3 6
-9 16
21 6
20 16


21 6
20 16
Flatten
21
6
20
16


21
6
20
16
%
fully connected